



Start It, Then Repeat

Ontario Math C3.1 — events + repeating

Name: _____

Date: _____

★ I can use an event to start a loop that repeats.

What starts the code?

Think of an alarm going off — that is what STARTS your morning. In code, an event block (like "when the green flag is clicked") starts a script.

After the event starts it, a repeat loop can run again and again. The event is the GO signal; the loop does the repeating. Bit will show you.

Read this script

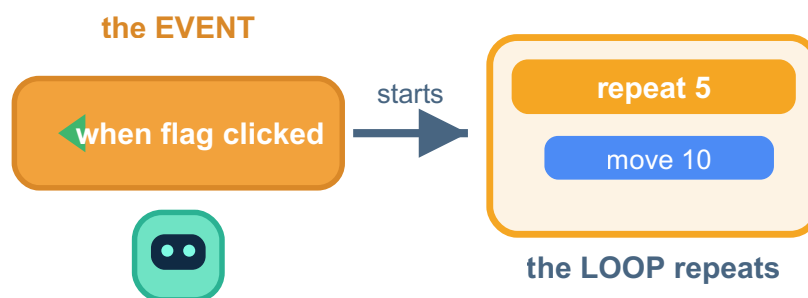
when this sprite clicked

repeat 3

next costume

wait 0.5 seconds

Clicking Bit is the **EVENT** that starts the script. Then the loop repeats the inside 3 times.



The event is the GO signal; the loop does the repeating.

1 Count it

In the script above, the event starts the loop. The loop says repeat 3. How many times do the inside blocks run? _____

2 Predict, then check

You change "when this sprite clicked" to "when green flag clicked". The loop is the same. PREDICT: what is different now — WHAT do you do to start it? Write your answer.

3 Write your own

Make Bit spin 4 times when the SPACE key is pressed. Fill in the blanks: when _____ key pressed → repeat _____ → (turn)

4 Challenge

Bit has: when flag clicked → repeat 5 → (move). If you click the green flag ONCE, how many moves? If you click it TWICE? Explain why.

Big idea

An event block STARTS the code. The loop inside it does the repeating. Fire the event again and the loop runs again.



Start It, Then Repeat

Quick facilitation notes & answer key

What this worksheet teaches

An event block starts a script, and a repeat loop inside does the repeating. Students count the loop, change the event, write their own event-and-loop, and reason about starting the event twice. No coding experience needed.

Before you start

No computer needed — the worksheet shows the blocks and what Bit does, so you can teach from the printout. Optional: a Scratch-style app to drag the blocks and watch Bit run — you only press the green flag.

How to lead it (about 30 minutes)

1. Show it (you do). The event (clicking Bit, or the green flag) starts the script; the repeat loop inside does the repeating.
2. Do one together. Exercise 1 — the inside blocks run 3 times.
3. Let them work. Students change the event (Ex 2), write an event-and-loop (Ex 3), and reason about two clicks (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — 3 times. Exercise 2 — now you start it by clicking the green flag instead of clicking Bit (the loop runs the same).

Exercise 3 — when SPACE key pressed → repeat 4. Exercise 4 (challenge) — one click = 5 moves; two clicks = 10 (the event starts the loop each time).

If students get stuck — and ways to adjust

Watch for: thinking the event changes how many times the loop runs. The event only starts the script; the repeat number sets the count.

Make it easier: point to the event on top, then the repeat inside.

Make it harder: ask how many moves three clicks of a repeat-5 loop give (15).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The child sees the event starts the script and the repeat sets the count, and reasons about firing it twice.